

Self-Detrimental Avoidance of Rest *

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Abstract

Conventional wisdom suggests that the best one can do to earn more is to work more. But recent evidence highlighting the productivity benefits of rest challenges this assumption. At the same time, little is known about the demand and opportunity cost of rest. In the context of an online labor market experiment in South Africa, we explore whether workers fully capitalize on the returns to rest by studying demand for and returns to resting. Our findings reveal that while mandated rest boosts productivity by 0.46 standard deviations, few workers choose to rest. We further allow for heterogeneity and investigate whether preferences about break preferences align with maximal earnings. Contrary to the notion of selection on returns, we find that workers who prefer to skip breaks could have earned more by resting. These results suggest a disconnect between perceived and actual returns and highlight the importance of understanding misperceptions around rest and their broader implications for productivity and economic decision-making.

Keywords: rest, productivity, workaholism, low-income workers, labor leisure trade-off, awareness, production functions, factor returns, demand, gig economy

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1 Introduction

In classical economics, leisure and output from production are related to each other through time spent not working being used for production or leisure (MODEL?). Labor is for production or generation of income; leisure is for consumption of said income. This suggests a tractable trade-off between how much time to spend working versus consuming.

However, recently long workdays or weeks have been met with skepticism, especially in higher-income countries. For example, proposals of four-day work weeks have emerged across several high-income countries, and research has drawn attention to the perils of working too much, with Nekoei, Sigurdsson, and Wehr estimating the cost of worker burnout to be 2.3% of labor income in Sweden (2024). Furthermore, evidence from economics shows that engaging in restful or health-conscious activities such as napping, exercising, or meditating can improve academic performance or work productivity (Bessone et al. 2021; Cappelen et al. 2017; Cassar, Fischer, and Valero 2022; Giuntella, Saccardo, and Sadoff 2022; Shreekumar and Vautrey 2024). These discussions all point to the notion that engaging in leisurely activities may be more than just consumption of leisure time. If leisure is also an input to labor, the trade-off between the two becomes more complex.¹

Although the existing work in economics has begun to highlight the benefits of self-care activities, these studies typically do not consider the demand for self-care activities, which includes the individuals' beliefs about opportunity costs of engaging in such behavior. The real impact of self-care interventions with voluntary uptake thus hinges on the alignment between perceptions and returns. A key next step is to understand and characterize demand for these activities. Our study aims to fill this gap by measuring the demand for rest and characterizing how demand is correlated with the true returns.

We conduct an online labor market experiment with 790 lower-income workers in South Africa. The experiment is designed around three objectives. First, we estimate the productivity returns to rest in our labor market setting by randomly assigning workers to different work schedules—with and without rest. Second, we characterize demand for rest by eliciting workers' choices over work versus rest time. Third, we examine whether worker choices about rest align with their returns, by testing whether workers' who exhibit low demand for break have lower returns to rest. Lastly, we explore the role of misperceptions of returns in explaining the misalignment between returns and demand for rest.

Our experimental design and analysis are guided by a conceptual framework that captures a richer notion of leisure that encompasses both non-pecuniary returns and pecuniary returns for workers. The rational benchmark would suggest that workers demand for breaks hinges on their overall utility, which is comprised of these pecuniary and non-pecuniary components.

In classical economic models, allocating time toward leisure—and away from “productive, output-generating” labor—has negative effects on income. We can test this hypothesis empirically. Moreover, if leisure is regarded as consumption, we can assume that it yields positive utility. This would suggest that if

1. This increase in complexity also applies to labor being an input in leisure as suggested in work on the psychosocial value of work by Hussam et al. (2022) and Macchi and Stalder (2024). Our setting allows us to abstract away from this channel as we focus on solo work.

contrary to classical economics, leisure has a positive production value, we would expect that individuals take breaks. The converse implies that if we do not observe individuals taking breaks, we would expect their earnings returns to resting be negative. If there exists heterogeneity in returns, individuals may still choose optimally based on private, idiosyncratic information. In that case, the same patterns as explained before would hold—only at the individual level.

To investigate these dynamics, our experimental design allows us to identify the demand for rest as well as the net financial returns to rest. Importantly, these financial returns include the opportunity cost of resting: foregone earnings from working more. By understanding the opportunity cost, we fully understand which costs and benefits an individual is facing when deciding between resting and working. Otherwise, correlations of gross benefits may miss important costs incurred while resting. By measuring both demand and returns, we can assess how causal returns align with workers’ demand for rest. This sheds light on whether workers are aware of how to optimally allocate their “inputs” in a way that maximizes earnings.

The experiment spans approximately one hour. It begins with a brief baseline survey, followed by the actual work period, and concludes with an endline survey. The main work period is subdivided into five separate, smaller periods (“segments”). First, all workers participate in a cognitively-demanding work task for two segments. The task is so attention-demanding that performance drops off quickly. Second, after these two segments, we elicit workers’ preference to either work or take a break during the third segment. The majority of workers are then randomly assigned to either a work or a break condition regardless of their choice for this third segment. This allows us to uncover the causal effect of assigning rest on productivity.² Participants who take a break listen to a brief, restorative audio script designed to promote rest. To ensure compliance, there is a small monetary reward conditional on attending to the audio script.³ Finally, all workers complete two more segments of the work task and then the main work session concludes. Afterwards, we collect measures of well-being and beliefs about earnings. In our experiment, we capture productivity through earnings per-segment worked and overall earnings. The latter measure accounts for the opportunity cost of taking a break by pricing in the forgone earnings while resting. Workers thus face a clear trade-off: taking a break might increase their earnings in the last two segments, but it comes at the cost of foregone earnings during the third segment.

We document three key findings around the alignment of returns and demand. First, we test the conjecture from classical economics that breaks have negative pecuniary returns, both in terms of “earnings per-segment worked” and in terms of “overall earnings” in our experiment. Contrary to the classical conjecture, randomly assigned rest significantly increases post-break productivity (in the fourth and fifth segments) for the average worker by 0.46 standard deviations ($p < 0.01$). The increase in productivity compensates for the loss of output during the break. Total work earnings from workers who worked in segment 3, 4, and 5 are the same

2. To incentivize truth-telling workers are told that picking their preferred option increases the odds of being assigned said schedule.

3. During breaks, the work task stops, and the software is programmed such that there is no way for individuals to continue working. A restorative audio script (non-sleep deep rest/ yoga nidra, explained in more detailed in Section 3) plays automatically. We instruct participants during the breaks to encourage genuine rest, as we want to avoid that participants look for other, less-restful occupations in the meantime.

as total work earnings from workers who rested in segment 3 and worked in segment 4 and 5 exclusively. That is, despite having one fewer work segment to earn income, workers who took breaks did not experience a reduction in total work earnings compared to those worked for all segments. Through the lens of our model, this suggests a positive average effect of breaks on earnings, indicating that workers should opt for rest to maximize utility.

Second, we can test if the positive sign of the pecuniary returns to rest aligns with demand. Contrary to the prediction of our model, we find that only 19% of workers opt to take a break despite the positive earnings returns. The remaining 81% of workers prefer to work during the third segment. Even when allowing for noisy decision making, this is substantially less than predicted by a model of rational decision-makers when returns to rest are positive. We find that break-skipping behavior is hard to predict with observable characteristics. However, individuals with a strong work ethic, lower household income, social desirability concerns, higher performance in earlier segments, and lower levels of fatigue were *less* likely to opt for rest.

Third, studying the average treatment effects may be masking selection based on returns. That is, individuals may face different pecuniary returns to rest, and *their* choices may be maximizing earnings *for them*. We would expect workers who avoid breaks to be experiencing negative pecuniary returns to rest. In the experiment, we find that workers pecuniary returns to rest are not correlated with their demand for rest. All workers experience a positive earnings boost in post-break segments. As before, the returns to breaks are large enough to make up for lost earnings. Although our sample lacks the statistical power to detect a significant difference in returns between the two subgroups, our point estimates suggest that those who chose to take a break may have benefited slightly more than those who did not. To explore the robustness of this finding, we further estimate quantile treatment effects and find little heterogeneity in treatment effects. To understand whether other dimensions of heterogeneity might be driving selection on returns, we also use Lasso but find little characteristics that predict positive returns. These robustness findings support the idea that workers who would have preferred to skip the break were not maximizing their earnings.

These findings raise questions about why demand for rest is uncorrelated with returns. Our conceptual framework rests on the assumption that individuals have positive non-pecuniary returns to rest. If this assumption does not hold, it may be rational for workers to choose working over resting if the non-pecuniary disutility derived from resting outweighs the earnings gains. In the context of our experiment, sources of non-pecuniary utility or disutility could be, for instance, cost of effort of working, joy of resting, self-image concerns about slacking off, joy of working, boredom while resting. On average, we find that overall self-reported happiness is the same across treatments. Additionally, we do not find any evidence of self-image concerns by comparing self-reported emotions felt around the time the random assignment decision was announced. If self-image concerns were to arise, we would expect workers who enjoy the break but don't choose the break because they fear that "indulging" in it is a sign of weakness, to feel relieved when they are "force assigned" the break against their choice. Instead, we document the opposite: workers are unhappy about being assigned the rest. When we survey workers at the end of the experiment about the reasons underlying their rest choices, most cite hoping to maximize their earnings. Very agree with the statement that they did not believe they would enjoy the break. We can repeat our previous analysis about the effect of randomly assigning rest on the latter subsample of workers of who did not choose a break albeit reporting it to be a

pleasant activity, and we again find a positive earnings effect of rest. This suggests these workers could have increased their overall utility by choosing to rest.

(STOPPED HERE) If the aversion of breaks is not driven by non-pecuniary elements in the production, an alternative explanation is misperceived financial returns to rest. These misperceived returns could lead to consistently suboptimal decision-making if the misperceptions are systematically deviating from the true returns. Using posterior data on incentivized earnings predictions of the two work schedules, we show that individuals systematically underestimate the realized earnings of both schedules. Causes of these misperceptions could be manifold including lack of information about one’s own performance decay, skill acquisition, and the benefits of rest or biased perception of returns due to stress.

Our results have significant implications for policy and future research. We posit there are many situations—outside of the context of our online labor market experiment—where the misalignment between the returns to rest and the demand for it is likely to arise as well. Specifically, we hypothesize that this misalignment may occur in environments where compensation is tied to meeting performance thresholds, and where fatigue quickly reduces the ability to sustain these thresholds. Mandating breaks is a potential policy worthy of discussion, however, the welfare impacts may vary across contexts and warrant careful consideration. Our experiment has documented this misalignment for gig economy workers who have flexibility over the schedules. Policy suggestions such as mandated breaks beg the question which views managers hold about the performance effects of rest. Given some views about the effects of rest for others are prone to the same biases.⁴ This highlights the need for additional research to examine the causes of misperceptions and what can be done to address them.

Our study is related to recent papers that use randomized controlled trials (RCTs) that document the positive returns to activities like sleeping, meditating, or exercising (Bessone et al. 2021; Cappelen et al. 2017; Cassar, Fischer, and Valero 2022; Giuntella, Saccardo, and Sadoff 2022; Shreekumar and Vautrey 2024). These studies utilize exogenous variation reveal the causal effect these activities. However, by assigning people to different conditions and/or incentivizing them to engage in activities, any costs that individuals may incur or perceive when engaging in these activities are masked. However, these opportunity costs are relevant in that they may explain low “voluntary” demand in real world contexts. Our study replicates the core finding of positive productivity returns to rest in an online labor market setting. Additionally, we can price the opportunity cost of rest, estimate net returns, and pair estimates these estimates with measures of demand for rest across workers. This is crucial to understand the impact of restful activities in the real world, which also hinges on uptake in contexts where the latter is voluntary.

Our study also connects to the broader theme of perceptions of production functions, specifically how factor inputs are best utilized. Previous research has explored this explicitly in agricultural settings (Bold et al. 2017; Hanna, Mullainathan, and Schwartzstein 2014) and most recently in the context of “allocative skills” among managers in labor markets (Caplin et al. 2023). Both strands of literature find limited awareness of optimal inputs usage. Unawareness of one’s own production performance is also echoed in low willingness to pay for productivity-enhancing, noise-protective gear (Dean 2024). Building on these insights, we focus

4. We discuss differences between choices for self versus others in Appendix ??.

on the labor-leisure trade-off in a context where workers can autonomously decide how much to invest in rest. We investigate factors influencing individuals' perceptions of a crucial component of "their own" production decisions: returns to investment in rest.

This dimension of working versus resting has been studied from the angle of procrastination and self-control, often focusing on real-effort provision and poverty (Augenblick, Niederle, and Sprenger 2015; Kaur, Kremer, and Mullainathan 2015; Schilbach 2019). Existing work has documented a tendency of workers to engage in breaks to procrastinate uncomfortable, effortful work tasks. We extend that work in the opposite direction, studying financially-inefficient overworking behavior among lower-income individuals.

Section 2 provides a conceptual framework for perceived returns and choices about rest. Section 3 outlines the experimental design that allows us to study demand for and returns to breaks. Section 4 discusses our main findings and section 5 considers potential mechanisms and external validity. Section 6 concludes.

2 Conceptual Framework

This section develops a conceptual framework to assess how individuals perceive and make choices regarding rest. Our focus is on understanding who opts to take a break and how these choices impact individuals' well-being. For the purpose of simplicity, we are agnostic about the exact underpinnings through which breaks influence individuals' work capabilities.

2.1 Returns to Rest

We consider a representative agent who derives their overall utility u from two different sources, pecuniary utility y and non-pecuniary utility v . An example of pecuniary utility could be earnings from a piece-rate wage. An example of non-pecuniary utility could be the joy derived from engaging in a pleasant restful activity or the mental cost of effort while working. Utility varies with the break variable b that captures whether the individual takes a break as part of their work period.

$$u_i(b) = v_i(b) + y_i(b)$$

For simplicity, we consider b to be binary. If $b = 0$ the individual works the entire work period without taking a break. If $b = 1$, the individual takes a break as part of their work period.

We define the causal effect of taking a break τ as the difference between the overall utility of these two states. And substituting the definition of u we can decompose the effects in the *earnings effect* Δy and the *non-pecuniary effect* Δv of a break.

$$\begin{aligned}
\tau &\equiv u(b = 1) - u(b = 0) \\
&= \Delta u \\
&= \underbrace{\Delta y}_{\text{earnings effect}} + \underbrace{\Delta v}_{\text{non-pecuniary effect}}
\end{aligned}$$

The canonical labor-leisure model assumes a trade-off between labor and leisure. Labor is defined as time spent working and producing output, which generates income that can be consumed; leisure is defined as time spent consuming. From this follows that more time is spent working, will lead to more output. Hence, in the classical model, resting is expected to reduce earnings because less time is spent “producing” and thus $\Delta y < 0$. Consumption of leisure inherently implies positive non-pecuniary utility, and $\Delta v \geq 0$.

Another way of interpreting $\Delta v \geq 0$ is that non-pecuniary utility may consist of many things, but on average, the net returns to resting will be positive. For instance, cost of effort of working and joy from resting might outweigh any losses from resting such as missing out on the psychosocial value of work or any self-image concerns about resting. Overall, this means there is negative non-pecuniary utility derived from working.⁵

Assumption 1: Non-negative Pecuniary Returns to Rest The non-pecuniary returns are non-negative:

$$\Delta v \geq 0$$

Hypothesis 1: Negative Earnings Effect of Rest

The pecuniary returns to rest are negative:

$$\Delta y < 0$$

2.2 Demand for Rest

Workers can make choices θ over whether they like to take a break b or not. For simplicity, the choice variable $\theta \in 0, 1$ is binary. $\theta = 0$ if a worker would like to work the entire period; $\theta = 1$ if a worker would prefer to take a break. The rational benchmark would suggest that a worker would like to take a break if the overall net returns on utility Δu are non-negative, and would avoid taking a break if the effect is negative.

$$\theta = \begin{cases} 1 & \text{if } \Delta u \geq 0 \\ 0 & \text{if } \Delta u < 0 \end{cases} \quad (1)$$

Equation 1 can be rearranged to capture the trade-off between pecuniary and non-pecuniary utility gains.

5. We acknowledge that $\Delta v \geq 0$ need not hold for every work task. In our experiment, we can shut down the psychosocial channel, and we discuss evidence that self-image concerns are unlikely to be driving our results in Section 5.

$$\begin{aligned}
\Delta u &\geq 0 \\
\Delta y + \Delta v &\geq 0 \\
\Delta v &\geq -\Delta y \\
\underbrace{v(b=1) - v(b=0)}_{\text{non-pecuniary utility gain of resting}} &\geq \underbrace{y(b=0) - y(b=1)}_{\text{financial gain of not resting}
\end{aligned} \tag{2}$$

Equation 2 implies that a worker will choose to take a break $\theta = 1$ if the non-pecuniary utility derived from resting outweighs the earnings gain of working. The latter is synonymous with the earnings loss of taking a break.

Consistent with Equation 1, we ask what we can learn about demand from observed pecuniary returns.

$$\theta = \begin{cases} 1 & \text{if } -\Delta y \leq \Delta v \\ 0 & \text{if } -\Delta y > \Delta v \end{cases}$$

Since $\Delta v \geq 0$ by Assumption 1, the sign of the pecuniary returns Δy will determine what we conclude about the demand for rest θ .

If Hypothesis 1 is satisfied such that $\Delta y < 0$, we would need to know how the magnitude of Δv compares to the magnitude of Δy . Since Δv is challenging to measure in the exact same currency as Δy , we cannot evaluate any observed choices about breaks.

In contrast, if Hypothesis 1 is violated and $\Delta y \geq 0$, individuals should demand breaks $\theta = 1$ in any case. The overall utility derived from resting will always be positive as both the pecuniary and non-pecuniary part are positive.

Hypothesis 2a: Optimal Demand for Rest

Non-negative pecuniary returns predict demand:

$$\Delta y \geq 0 \Rightarrow \theta = 1$$

Next we ask about the converse: What can we learn from demand about the returns to rest? If we see a worker choosing a break, we cannot conclude much about the pecuniary returns Δy . Large non-pecuniary returns Δv could cancel out smaller negative pecuniary returns Δy . Similar to before, we would need to measure the exact magnitude of the inobservable Δv , which is challenging.

However, if we see a worker not choosing a break, we know that they must feel that their overall utility effect Δu is negative. Under Assumption 1 that $v \geq 0$, this implies a negative earnings effect $\Delta y < 0$.

Hypothesis 2b: Optimal Demand for Rest

No demand for rest implies negative pecuniary returns to rest

$$\theta = 0 \rightarrow \Delta y < 0$$

Analysis of the aggregate data may be misleading if returns are heterogeneous. Individuals may still be demanding breaks optimally, but may select into rest according to private information about idiosyncratic returns. If we relax the assumption of a representative agent, Hypothesis 2 still holds for groups of individuals.

2.3 Potential Explanations of Suboptimal Demand for Rest

If we do not observe optimal rest-taking demand according to Hypothesis 2, this implies that individuals must *feel* like their overall returns to resting are different than we observe. This misalignment according to Hypothesis 2 can have two not-mutually-exclusive causes. We will consider an example where the demand for rest is negative, but the pecuniary returns to resting are positive. If an individual chooses not to rest, this must be because they believe the overall utility effect rest to be negative. If a violation of Hypothesis 1 is empirically documented and the pecuniary returns to resting are positive $\Delta y \geq 0$, the break taking condition suggests that either non-pecuniary utility must be negative or individuals must hold wrong beliefs about the pecuniary returns to rest.

First, Assumption 1, which states that non-pecuniary effects of rest are non-negative, need not hold. If $\Delta v < 0$, we can no longer claim that a positive earnings effect of rest implies positive demand for rest $\theta = 1$. We provide empirical proxies in favor of the positive non-pecuniary utility assumption in Section 5.

Second, individuals may hold different perceptions of the pecuniary effects Δy . We denote these (mis-)perceptions of variables using a tilde.

For example, if individuals view the earnings effect of rest to be large and negative it may be rational to avoid resting from their perspective. The choice of a break thus depends on individuals' *perceived* returns to rest, and equation 1 turns into equation ??.

$$\theta = \begin{cases} 1 & \text{if } -\widetilde{\Delta y} \leq \Delta v \\ 0 & \text{if } -\widetilde{\Delta y} > \Delta v \end{cases}$$

Thus, misperceived financial returns to rest can explain a misalignment between demand for rest and its returns.

Hypothesis 3: Pecuniary Returns to Rest are Underestimated

If demand for breaks is negative but pecuniary returns are positive, individuals misperceive the pecuniary effects to be negative.

$$\theta = 1, \Delta y < 0 \Rightarrow \widetilde{\Delta y} < 0$$

To summarize, our model provides us with the following insights and hypotheses.

1. The randomized assignment of breaks b allows us to sign the causal effect of rest on earnings y .
2. Under the assumption that non-pecuniary returns to resting are positive, we can test whether non-negative pecuniary returns imply positive demand for rest.
3. Considering heterogeneity in returns, we can test if negative demand for rest predicts negative pecuniary returns to rest.

4. If demand is negative albeit pecuniary returns are positive, we can test whether returns to resting are systematically underestimated.

3 Experimental Design

3.1 Experimental Design

To study the impact of and demand for rest, we recruit low-income workers from South Africa to perform an attention-demanding real-effort task for a work period that lasted approximately one hour. The work period was split into five segments, each approximately 10 minutes in duration on average.⁶ At a high level, participants first work for two segments, then face a choice about the third segment, and then work for another two segments. Afterwards, we collect additional measures on well-being and beliefs.

Our design around the choice is similar to the selective trials framework described in Chassang et al. (2012). We use a simplified binary version. All workers begin by working two segments. This allows them to get experience with the work task. Participants then express a preference over taking a break or working in the third segment. We tell participants that choosing their preferred option will increase the odds of receiving this option. We then implement workers' choices for a randomly selected subset of 5% of the participant pool.⁷ The remaining participants (95%) were randomly assigned with equal probability to either take a break or to work during the 3rd segment.⁸ After the third segment, in which participants are either working or taking a break, everyone works for the remaining two segments. At the end of the five segments, subjects answer a brief endline survey about their beliefs and work period experience.

Figure 1 captures the design, where θ represents preferences over breaks and b represents break realizations.

The real-effort task that subjects engage in during the work period is designed to mimic an air traffic control software, using the same images and general task structure as in Waldfogel et al. 2019. The task is also known as a “sustained attention release task” (SART) or “go no-go” task in the psychology literature.⁹ We use this task as it requires constant attention and is very cognitively-demanding without requiring any special skills or expertise. The constant attention always implies that we can observe at the 2-second level if participants doing the task or doing something else. Since attention is a finite resource in the short-term, this task also has an automatic performance decay built in.

During the task, workers are shown a sequence of images, each depicting two flight trajectories. For each image, workers then decide whether the two flight trajectories shown are “safe” (meaning the flights are

6. Segment length varied due to lags in the Qualtrics server and Internet speed. To account for these lags, we set the overall time constraint of 65 minutes based on a conservative estimate of the time it would take to finish the work period (from piloting). We also accounted for this at the beginning of the study by telling subjects upfront that the exact time of the study varies and cannot be influenced by them.

7. The randomization was implemented via Qualtrics.

8. To address concerns of behavioral responses to not receiving their first choice, we first inform everyone who was part of the 95% randomly allocated subset that they were randomly assigned to a treatment by the computer. This is meant to reduce behavioral differences in frustration between people who received their preferred outcome and those who did not.

9. SART tasks have also been used in economic studies of cognition and human capital (Brown et al. 2022).

not about to collide) or “not safe” (meaning the flights are about to collide).¹⁰ For every image, they have two seconds to decide whether it is safe or not-safe. If it is safe, workers are instructed to click the “safe” button. If it is unsafe, they are instructed to *not* click the “safe” button and instead let the image lapse after the two-second timer is up.

For each work segment, there are 300 images to evaluate. Of those, 280 are safe and 20 are not safe. Only one image is shown at a time, and the order of safe and not-safe is random. The disproportionately large share of safe images makes it easy to get into a habit of constantly clicking, which is part of the original intention for these types of attention-demanding tasks. The idea is that this imbalance in image types and intended response behaviors requires constantly paying attention to withhold the impulse to click on “not safe” flight trajectories. This in return temporarily uses up the finite attention resource and leads to decays in performance.¹¹

Subjects are compensated with a guaranteed £8 base pay for a 75-minute experiment for participating in the study. Additionally, subjects can earn bonus payment for successful completion of work segments. In the first three segments, bonus payment is £1 per successfully completed segment. In the last two segments, bonus pay increases to £3 per successfully completed segment. The pay schedule is summarized in Figure ??.

Higher earnings in the last two segments serve the purpose of increasing the salience that it is important to be attentive and perform well toward the end. This should guide participants toward making optimal choices about resting and working and might highlight that resting can have positive productivity effects. These increasing incentives are comparable to low-stakes midterms and high-stakes final exams or to surge prices that Uber drivers might face at night.

A segment is considered successfully completed if all criteria are satisfied: (i) at most 2 not-safe flights are missed, (ii) more than 75% of all flights are correctly identified, and (iii) it is completed within the work time window of 65 minutes. The first constraint is what makes the task challenging. It forces workers to pay attention to these not-safe flights that are rare and easy to miss by always clicking out of habit. This is the key margin along which performance differs across subjects over time. The second constraint is a basic attendance constraint. We aim to ensure that individuals are actively engaging with the task throughout the work segment. If we would not enforce some measure of getting both types of flights correct, workers could game the task by always clicking or never clicking. The third constraint prevents individuals from pausing the task on their own and equalizes working conditions across subjects.

By choosing an all-or-nothing payment structure, we mimic piece-rates that are common in lower-income work settings. It also simplifies the trade-off between taking a break and working for the subject. Thus the opportunity cost of a 10-minute break is well-defined and lessens the burden on the worker in terms of carefully thinking about how many images they expect to get correct, which would be the case if workers would be paid in pennies for every correctly identified flight. It also makes it easier to create a bundle of critical and non-critical flights, which allows us to ensure attentiveness of the workers as mentioned before.

Participants who take a break earn a flat payment of £0.5. They earn this payment as long as they active

10. Example images of a safe and a not safe flight trajectories are in Figure ?? in the Appendix.

11. These performance decays have been documented in psychological studies using these sustained attention release tasks.

engaging with the break that we “provide”. During the break, the work task is inaccessible, and the program auto-plays an audio script, which subjects are asked to listen to. The audio script focuses on certain types of breathing, also known as yoga nidra or non-sleep deep rest that have been shown to have calming effects (Balban et al. 2023).¹² We check compliance with the audio-scripted break by asking one question at the end of the experiment that asks about the general content of the audio script. When facing the choice, workers know about the potential break payment as well its contingency on attending to the break.¹³

Workers do not receive feedback on their performance through the task. This applies to both the performance in the main work period and attending to the audio script during the break. Thus the focus lies on workers understanding their own performance of their “production function,” as a result of their factor inputs.

To ensure a high-data-quality sample of online workers, individuals had to successfully complete an online training and correctly answer questions about the instructions during the training in order to be eligible to be hired for the main work period. Training included a description of the incentive scheme, instructions for the task, and a preview of the break choice they would have to make during the study. We also collected basic demographics, including gender, state, education, employment, and language, as well as unincentivized measures of baseline initial task performance during this training.¹⁴ This also gave individuals a chance to practice and gain experience with the task before they decided on whether to participate in the full work study. We select lower-income individuals with a per-person income of less than ZAR 6,000 per month.¹⁵

Participants who passed the screener are then invited to the main study. The invitation goes out a couple of hours after participants completed the training, and participants could take the study at a time of their choosing as long as it was within one sitting.

Before the work period starts, participants respond to questions from the Marlow-Crowne Social Desirability Scale (Reynolds 1982; Dhar, Jain, and Jayachandran 2022) and the World Value Survey about work ethic¹⁶. During the work period, we observe performance (in the form of mistakes, which translate into bonus payments), self-reported fatigue after segment 1 and 5, as well as individuals’ preferences for breaks (for themselves and for others). After the work period, we elicit individuals’ beliefs about the number of mistakes in the last segment as well as self-reported happiness and task satisfaction.

We also ask for incentivized predicted earnings by treatment . This provides us with a measure of perceived returns to rest. Lastly, we collect qualitative data about their break choice and engagement with the break audio script.

12. We use a script from the website nsdr.co from a point in time when the website and its audio scripts were still free.

13. Workers know that they will be asked questions about their experience, but we keep the details vague to avoid that workers can game the system.

14. As part of this task training, participants first practice with feedback. Afterwards, they practice without feedback. The latter allows them to experience what the work task will be like.

15. For comparison, senior citizens with monthly income below ZAR 6550 are eligible for social assistance from the government. We focus on this demographic as we are interested in individuals who are financially constrained and for whom this online survey income represent a significant source of income. Some of these individuals also work in gig-economy type settings where they do image labeling tasks similar to our air traffic control task.

16. We choose item 39-41 from the World Value Survey, Wave 7 (Haerpfer et al. 2022) and elicit these items alongside a multitude of questions to minimize experimenter demand bias

The South African online setting lends itself well for this experiments as online gig work is very prevalent in lower-income countries. Workers on cloud gig platforms may engage in tasks that are similar to our air traffic control task e.g., image labeling for AI companies. Moreover, we believe that our results are relevant for gig economy workers who have autonomy over their schedules. Thus the decision whether to work one additional segment or whether to take a break translates well to the context of accepting another “human intelligence task” (HIT) or taking a short break.

In sum, the collected data will allow us to understand, first, how breaks affect productivity, defined as average earnings per segment worked, and well-being by comparing the 95% of the sample that were randomly assigned. Second, data on spontaneous preferences for a break provide information about the demand for breaks. Third, we can uncover the treatment effect on the treated and untreated (TOT, TUT) by comparing individuals assigned to breaks or no breaks within those who preferred a break or did not prefer a break.

3.2 Sampling Procedures

We invited N=1,577 English-speaking participants in South Africa to participate in an initial training session. This training functions as a screener. Participants know that upon successful completion they may be invited to participate in the 75-minute long main study. Among all those who were interested, 83% (N=1,313) subjects passed our screener and were eligible for the study. The 10-minute screener consists of basic demographic questions, the instructions of the experiment a task trial period with feedback (to give participants an opportunity to learn), a task trial period without feedback (to mimic the real study), and a quiz about the instructions including the incentive structure and the break as well as an audio sound check (to ensure the audio script could be played during the break). Of those who passed, N=1,035 people met our “lower-income” criteria.¹⁷ We invite all these online workers to participate in the main study, and N=XXXX accept our work invitation. For our analysis, we restrict the sample to participants who only took the survey once, and N=906 unique participants follow the invitation.¹⁸ Of those, 834 participants finish all segments of the work task. We focus on this sample for our main analysis. Of those who completed the work period, 826 finish the entire study including the post-experiment survey questions. Finally, since we only randomly assign 95% of all observations, we restrict our analysis to those N=790 randomly-assigned participants out of the 834 who completed the work session and post-session survey.

17. Although there is no way for us to verify self-reported income, it was unlikely that participants were aware of our income selection criteria, since the income questions appeared among many other demographic questions. Income was elicited in categories, which we then top coded and divided by the number of household residents. If this imputed value was below ZAR 6,000 we invited household to participate. Households knew from the beginning that not everyone who passed the training would be able to participate, but they were not informed explicitly about the income eligibility criterion.

18. Some participants took the study multiple times—even though the instructions clearly stated that this was not allowed and would not be remunerated. We cannot distinguish between participants facing technological issues and restarting (exogenous shocks) and participants who would like to have a better second attempt and stop and restart the survey (endogenous). So we discard all these observations for our analysis. Participants were rewarded for their time nonetheless.

4 Results

4.1 First Stage: Break Treatment Compliance

Participants assigned to take a break were instructed to listen to an audio script. To ensure compliance and prevent survey hopping, we told workers upfront that they would be asked incentivized questions about the break experience at the end of the work period. At the end of the study, 94% of participants reported attending to the audio script. Of those, 68% correctly answered a question about the breathing technique used in the audio script. Among the remaining 32% who incorrectly answer the question, 15% of individuals admitted to engaging in other activities during the break ($N=35$), but 85% still claimed to have listened to the script despite missing the question ($N=XX$). Importantly, having an initial preference in favor of (or against) a break was not correlated with higher audio script compliance.

4.2 Effects on Productivity and Earnings

To understand the isolated benefit of rest in the absence of costs, we compare the effect of randomly assigning breaks on post-break earnings. Any differences between the two groups may reflect a mix of two dimensions: first, increased focus due the restorative audio script, and second, the lower focus due to accumulated fatigue from not resting. Figure 3 shows that individuals who were assigned a break are more likely to earn a bonus payment in segments 4 and 5. Table 2 show that this effect is statistically significant. We use an ANCOVA specification and control for baseline performance in Segments 1 and 2 before the randomization took place in Column 1.¹⁹ Additionally, we control for gender, education, employment status, year of birth, province, and per-capita household income in Column 2.²⁰ The latter column is our preferred specification and shows that individuals can on average earn £0.36 more per each segment after the break holding past performance and individual characteristics constant. Thus sums up to £0.72 higher earnings in Segment 4 and 5 combined if workers were assigned to rest in the third segment. These results are statistically significant at the 1% level. Our positive results on the productivity-enhancing effect of short restorative breaks complement the existing literature around restful activities, which has documented that incentivizing people to engage in behaviors like sleeping can have positive productivity effects, as reflected in outcomes like improved academic grades (Giuntella, Saccardo, and Sadoff 2022).

We next quantify how the increased post-break productivity compares to the cost of forgone earnings incurred while resting. That is, are the higher earnings in Segment 4 and 5 due to increased focus or lower fatigue among “break takers” worth the lack of earnings in Segment 3? The first two columns of Table 3 regress work earnings in Segment 3 through 5 on the random assignment of a break. These work earnings only include bonus payments from completing work segments. They do not include any payments from attending to the break. We focus on Segments 3 to 5 as Segments 1 and 2 took place before the randomization.

19. We discuss the ANCOVA specification and differences in baseline performance in Appendix ??.

20. This and all following computations are restricted to the 95% sample of workers who were randomly assigned to one of the two work schedules. The 5% of workers whose choices were implemented are discarded from any analyses. We further limit the sample to individuals that completed the entire work segment and have no duplicate entries.

We find null effects on work earnings across these last three segments, implying that workers break even when taking a break. That is, total work earnings were the same even though workers assigned to the break group only worked for two segments and thus could earn at most £6 from working during segment 3 through 5, while workers in the assigned group could earn £7 from working during segment 3 through 5. This demonstrates that in the context of our experiment, breaks pay for themselves and workers are just as well off monetarily with taking a break as without. The effects of magnitude £0.02 are not precisely estimated but very close to zero with standard errors of approximately £0.15. The mean work earnings from segment 3 through 5 in the no-break group are £3.8, however, a comparison of means is not informative as treatment assignment is unbalanced in baseline performance.

These first two columns of work earnings focus on “output” and do not include the small payment earned by workers for taking a break and answering the break attendance question correctly. In Columns 3 and 4 ³ we consider overall earnings including the break payment. From the perspective of a worker in our study, this small compliance payment that can be earned during a break is also payoff relevant. We see statistically significant higher earnings for those who were randomly assigned the break than for those who were assigned to work during that segment. On average, workers earned £0.35 more in Segment 3-5 if they were assigned a break ($p < 0.05$).²¹ Hence, a worker who would like to maximize their earnings should have chosen to take the break in this setting.

In Appendix ??, we verify that our earnings results are not mechanically driven by payment schemes and robust to relaxing the 65-minute constraint that workers faced for completing the work segments. To complement our main analysis using earnings as the key outcome, we also analyze mistakes as the measure of productivity in Appendix ??.

4.3 Demand for Breaks

Despite the fact that workers earn significantly more under the schedule with a break, demand for breaks is low in our experiment. Only 19% of subjects prefer to take a break when given the choice. The preference for a break is elicited after finishing the first two work segments and immediately before the implementation of the break, with the goal of eliciting the choice when workers have experience with the task and have build up some fatigue up until that point. When making their choice, workers know the full earnings scheme for both work schedules, including the fact that later segments will pay higher bonuses (which we validate explicitly in the attention checks). The low take-up of breaks contradicts our Hypothesis 2a that breaks would be preferred if benefits are positive.

Figure 4 displays coefficient estimates from a regression of break preference on individual characteristics including demographic variables, values, and past performance. Two of the World Value Survey (WVS) questions meaningfully predicts workers’ preferences to take breaks. On average, agreeing one point more with the statements on “work being a duty to society” and “work should always come first” (on a Likert scale from 1 to 5) is associated with a 3pp and 4pp lower propensity to demand a break. We also find that lower work performance in the baseline period significantly predicts higher demand for breaks. We find that higher

21. Implications for earnings gains during the work day calculations are discussed in Appendix ??.

self-reported fatigue after the first segment predicts a higher likelihood of preferring a break. We discuss mechanisms that may underlie these results in Section 5.

4.4 Do Choices Over Breaks Maximize Earnings?

The previous regressions pooled all workers and analyzed both returns and choices on average. However, an analysis of the aggregate data could be misleading if there is heterogeneity in returns and if individuals select into rest based on private information about idiosyncratic returns. Hypothesis 2b predicts that individuals who chose to work as opposed to resting experience negative pecuniary returns to rest. We test this hypothesis by splitting our sample into two groups based on workers' choices about rest and repeat the previous analysis. Qualitatively, we see some evidence in Table 4 that workers benefit from breaks regardless of their initial choice. Column 1 and 2 focus on workers who did *not* choose the break. As shown in column 2, they earn £0.34 more if assigned the break holding initial performance and demographics constant. The effect is marginally significant at the 10% level. Column 3 and 4 show the same results for workers who chose a break. The point estimate of £0.4 is large, albeit not significant. However, we cannot rule out that we were underpowered to detect effects in the small sample of $N=149$ individuals who chose a break and who supplied demographic characteristics. Columns 5 and 6 pool both groups and include an interaction term for the initial choice. We again see that the random assignment of breaks significantly increases post-break earnings of workers by £0.32 at the 10% level. The interaction between a preference for a break and being assigned a break is not significant, which may reflect low power as discussed before. Under the assumption that individuals aim to maximize their earnings and there is no negative non-pecuniary utility from resting, we can conclude that their choices to avoid breaks are not maximizing their utility.

We complement this exercise by testing whether there might be different dimensions of heterogeneity that may matter. We use Lasso to predict whether any demographics can predict the gain in productivity from rest and find that few variables do so. Furthermore, we compute quantile treatment effects and find no evidence of heterogeneous effects of breaks, as the differences between quantiles are statistically insignificant. Taken together, our results suggest that there may be similar biological responses to fatigue and rest.

5 Why Is the Demand for Rest Low?

The results in Section 4 highlight the contrast between positive returns to rest and low uptake of rest. Per our framework from Section 2, demand for breaks depends on the overall utility effect of rest, which encompasses pecuniary and non-pecuniary utility. To rationalize a preference for working instead of taking a break, individuals must believe their overall return to taking a break is negative. This could happen through two non-exclusive channels that we will discuss below. First, our initial assumption about the sign of non-pecuniary returns to rest may be wrong. Second, individuals may misperceive the pecuniary returns to resting.

5.0.1 Negative Non-Pecuniary Utility from Rest

As stated in Section 2, we assume that the non-pecuniary returns to resting are positive. Examples of this include a positive cost of effort i.e. expending energy while working is costly and thus individuals derive a positive utility benefit from not working because they do not have to pay this cost. Another option is that workers may innately enjoy resting e.g., they may derive utility from closing their eyes and refocusing, or outside the context of our experiment, they may enjoy a break where they drink coffee or go for a walk.

However, non-pecuniary utility effects of resting need not be positive. First, one could imagine that individuals also enjoy working, e.g., because they derive psychosocial benefits from working (Hussam et al. 2022; Macchi and Stalder 2024), because they enjoy “getting things done” or solving a complex task, or because they derive meaning from another dimension like helping others (**MeaningAtWork**). Second, they may not enjoy chatting with colleagues during a coffee break or may find a non-sleep deep rest break boring. Lastly, there may self-image concerns. Third, while a worker might not enjoy the task and might enjoy resting more, they might derive non-pecuniary utility from signaling to themselves that they are hard working, that they are not slacking off, that they are stronger than people who need to rest, and that they are capable of using every minute of the day productively.

Contrary to the first argument, we believe that it is unlikely in our context that individuals derive positive non-pecuniary utility from the task itself. There is no social dimension, thus shutting down the psychosocial channel. Since no feedback is provided about performance and the task is quite repetitive, we also believe it is unlikely that participants derive any innate utility from that. Lastly, while it is framed as an air traffic control task, which may be a meaningful service task, the lack of feedback and concreteness about the job makes it unlikely that people find the work meaningful.

Second, if individuals truly do not enjoy the content of the break that is provided by them, but are aware that not working increases their earnings, they would be free to engage in a different form of a break. Additionally, it is unlikely that relative to completing a repetitive task, an individual would find closing their eyes and taking a couple minutes to restore less pleasant. Among individuals who did not choose the break, 71% disagreed that they did so because they thought they would not enjoy the break.

Lastly, if self-image concerns would drive these results, we would expect that workers feel secretly relieved about being assigned a break—because they get to enjoy a break without having to actively choose it. To understand this dimension, we collected data at the end of the experiment where we ask participants to reflect back on how they felt when they learned about their random assignment and rate it on an emoji scale. Only 25% of participants who did not want to take a break report that they felt happy when they first found out about their assignment. 44% felt reported feeling unhappy, and the remainder did not report any strong views.

These hypotheses about no negative effects of rest on overall utility are also reflected in self-reported measurements of well-being. At the end of the experiment, we ask individuals to self-report whether they feel rather happy or rather unhappy at the moment. This variable is rather coarse and may encompass both pecuniary and non-pecuniary utility—although people did not know their earnings by the time that they answered this question. In either case, as shown in Column 1 of Table ??, we do not uncover any statistically

significant differences between groups. This suggests that breaks do not induce any disutility. Furthermore, after segment 1 in the experiment (before randomization) and after segment 5 (after the experiment), we ask people to report if they feel “rather tired” or “rather alert.” We find that workers assigned to take a break are significantly less likely to report feeling tired holding constant levels of baseline fatigue.

As we allowed for heterogeneity in pecuniary returns to rest, there might be heterogeneity in non-pecuniary returns. For a subset of workers who do not enjoy working less (e.g., because they do not enjoy listening to audio scripts or sitting still), the breaks can indeed represent a suboptimal choice if the pecuniary utility they derive from higher earnings is less than the non-pecuniary utility cost from resting. However, measuring both forms of utility in the same currency can be challenging. Thus, we cannot confidently ascertain whether individuals who do not enjoy breaks would be better or worse off by taking a break. However, we can focus on a subgroup of workers that disagree with the statement “I did not think I would enjoy the break.” Approximately 71% of the participants who choose to work believed they would have enjoyed the break. We collect this qualitative data at the end of the experiment when we ask individuals to explain their choices in favor of or against breaks. We believe that for this group the positive non-pecuniary returns assumption is valid and as a robustness check, we repeat the previous exercise about the effect of randomly assigning a break among this subsample. Table ?? suggests that this subsample also would have earned more by resting. Thus, the data suggest that these workers’ choices were mistakes.

5.0.2 Misperceived Financial Returns to Rest

The previous section suggests that choices are unlikely to be driven by negative non-pecuniary utility derived from rest. Another option is that even if pecuniary returns are positive, individuals perceive them to be negative. In that case, it may be a rational action for them choose to avoid rest given their beliefs. This hypothesis is also in line with qualitative survey evidence, which also suggests that individuals’ primarily chose breaks to maximize their earnings.²²

In our model, a worker’s choice about whether to take a break depends on the perceived net returns $\tilde{\tau}$: the worker takes a break if and only if the perceived return is positive. The total return comprises two parts: pecuniary returns, which capture the monetary effects of earning more in post-rest work periods, and non-pecuniary returns, which capture any experienced utility of the break versus the work task such as joy or guilt.

After the completion of the work period, we elicit incentivized predictions from each group to predict the average earnings of each work schedule. These allow us to understand (i) how posterior perceived returns correlate with choices, (ii) if there was a treatment effect of being assigned a break on predicted earnings, and (iii) to what degree the predictions of the treatment effects are correct in levels.

22. First, qualitatively, the factor most frequently cited by workers as driving their choice was earnings considerations. 46% of workers who chose to forgo the break mentioned that not taking a break would maximize their earnings during the work period, whereas other factors were mentioned much less frequently (33% preferred choosing their break schedule for themselves; 21% reported not liking to take breaks, 17% did not wish to be seen as lazy, and 17% were skeptical of engaging in the NSDR script during the break). Second, quantitatively, we find that beliefs about the earnings return to breaks is strongly correlated with and predictive of break choice.

First, we find that these posteriors beliefs strongly reflect individuals' performance for break choices. On average, individuals who choose a break perceived earnings to be X high without a break. Individuals who did not choose a break perceived earnings to be higher when working. Second, we find only a weak treatment effect of being assigned a break on perceived earnings. We cannot reject the hypotheses that the beliefs are different between groups. Third, we find that the pooled perceived treatment effect is significantly lower than from the true effect. This implies that perceived treatment effects were significantly biased and underestimating true returns. We now discuss several factors that may have contributed to the formation of misperceptions.

Lack of Familiarity with Work Task and Inawareness of Productivity Decline: A potential explanation for the low take-up of breaks is the lack of information about the nature of work task – in particular, workers' inattention to their own performance decline. While our current design cannot disentangle misperceptions about the future productivity returns to rest from misperceptions about past productivity, we argue this is unlikely to be driving the low demand for rest. First, workers make their choice of whether to take a break after already having completed nearly half of the work period: by the time they face the choice, they have already worked for two full segments for approximately 20 minutes. Even over this time frame, workers face declines in their performance of X%. Furthermore, workers are aware of this decline – when we ask them to guess the number of mistakes made on non-safe flights for each segment, they report making Y% more mistakes in segment 2 than segment 1. In sum, workers not only experienced but also were aware of the performance decline at least directionally.

Overweighting of Skill Aquisition with Practice: Another motivation that could explain relative misperceptions of the benefits of resting and working is the idea that workers may have wanted to get more practice with the work task before reaching the higher-bonus last two segments, since the bonus pay was \$3 per segment for the last two segments while it was only \$1 for the first three segments. If the returns to practice are high enough to outweigh the effects of fatigue, workers may want to use the third segment for additional practice to ensure that performance is high for the highest-paying segments. However, as documented by our results, the net returns to resting and starting refreshed outweigh any potential gains from additional practice. Participants also did not explicitly mention practice or learning for later sessions in the open text response question that asked about reasons for their choice. We provide evidence of learning not mattering in Appendix ?? That said, we cannot rule out though that participants could still be holding this view, considered it as "maximizing earnings when replying to our qualitative questions", and that this is driving break-taking behavior.

Lack of Familiarity with Rest Benefits: Similar to the idea that participants may not be familiar with the task and its cognitive burden/ the decline in performance, they may also not be familiar with the restorative potential of taking a break. Our ITT effect estimates the benefits of non-sleep deep rest breaks. As argues in Appendix ??, we consider this to be a lower bound on the potential effects of breaks as people might otherwise freely choose activities that are even better for them. Additionally, there is always an option to “just” rest, which is something that participants should be familiar with. Only 17% of individuals state that skepticism about the audio-script was one of the reasons why they did not choose a break.

Stress: An emerging literature has studied the implications of stress on decision-making. Literature from

neuroscience has shown that stress can alter our perceptions and may lead to decision-making based on heuristics (CITE). Intuitively, one can imagine that the choice of whether to go for a run on the morning of a big deadline is challenging. The run might make it easier to focus and lead to work getting done faster or at a higher accuracy. On the other hand it takes up time. Similarly, if one is financially constrained, one may feel compelled to work more to earn more. In the context of this experiment, we hypothesize that temporal, and in particular, financial stress may lead to suboptimal decision-making. Concerning financial stress, Figure ?? documents that income is negatively correlated with the propensity to take a break.

The psychology literature suggests that stimuli may be more noisily perceived under time pressure (CITE–stopped here).

If pecuniary returns to leisure are more noisily perceived than pecuniary returns to working (e.g., because there is a posted wage), it is plausible that people will hold more misperceptions about returns to rest than returns to working. However, unless there is a bias, we would expect the misperceptions and errors to go in both directions since one can under and overperceive. If there is a bias such that people systematically underestimate the returns to resting relative to working this could explain why rest is chosen less than optimally. It is conceivable that determining the returns to resting feels complex or effortful or time-consuming and that is why people do not like to engage in this estimation and prefer to work.

6 Conclusions

This study investigates the impact of taking breaks on productivity and overall earnings in a controlled experimental setting. Our findings reveal several key insights into the role of breaks in enhancing worker performance and well-being.

Firstly, we find strong evidence that taking a break positively affects productivity. Participants who were randomly assigned to take a break exhibited a significant increase in per-segment earnings post-break. This productivity boost is robust even after controlling for baseline performance and demographic characteristics, suggesting that breaks can effectively rejuvenate workers and improve their output.

Secondly, when examining overall earnings, our results indicate that individuals who took a break earned comparable or slightly higher total earnings compared to those who did not take a break. This finding suggests that the productivity gains from taking a break offset the opportunity cost of the time spent resting. Even after accounting for baseline performance and demographic variables, the positive effect of breaks on overall earnings persists, highlighting the economic value of incorporating rest periods into work schedules.

Despite these benefits, the demand for breaks among participants was notably low, with only 19% expressing a preference for taking a break. This low uptake may be attributed to a variety of factors, including individual perceptions of their ability to perform tasks, beliefs about the opportunity cost of rest, and varying levels of fatigue. Participants who reported higher fatigue or lower initial performance were more likely to opt for a break, suggesting that individual needs and conditions play a significant role in the decision to take a break.

Our analysis of whether participants' choices over breaks maximize their earnings yields nuanced insights. Since breaks appear to generally enhance productivity and overall earnings, those who chose not to

take a break did not maximize their earnings potential. This result is corroborated by low heterogeneity in treatment effects.

The strong treatment effects across the entire sample have important implications for policy-making because it suggests that mandated breaks may be beneficial for both workers and firms. Further research should investigate the reception of these mandates.

Additionally, our results raise questions about the sources of the suboptimal demand for rest. Assuming that breaks do not induce a strong disutility, misperceived returns to rest are the most likely source of low demand. We find preliminary evidence of misperceptions by studying differential updating with respect to posterior returns about the effects of breaks across treatment groups. Such initial misperceptions could stem from for example a sheer lack of information about the positive returns to rest, systematic overconfidence about one's ability to perform well, scarcity of cognitive resources and poor decision-making under financial stress, or feeling guilty about resting. Further research could explore these avenues along with investigating how individuals best learn about their "own" production function over time.

6.1 Moved: External Validity and Relevance

6.1.1 Relevance of Features of the Experimental Design

We document misalignment between returns to rest and demand for rest in a setting where returns are positive. The positive financial returns hinge on a quick performance decay that is "outperformed" by the recovery during the rest period. While there may be performance decay in any context, a specific aspect in this setting is that the decay also directly maps into lower earnings. For example, if the threshold were such that the decay would still be above the pay threshold, there were no additional benefit of rest. This suggests that performance decay in cognitively demanding tasks is likely biologically driven—which may also explain little heterogeneity in the benefits of rest. That said, the monetary trade-off will always be context specific. It is plausible that this misalignment between demand and returns arises in similar contexts of high returns to restful activities. More research is needed to distinguish between misperceptions that are specific to the concept of resting and misperceptions about one's own "production function" more broadly.

6.1.2 Scalability and Policy Implication

Our results shed light on the behavior by independent workers. This has relevant implications for self-employed workers or those in gig economy type of employments who have autonomy over their schedule. In deciding how to spend their time, they constantly actively trade off one decision versus the other. Given that that may be payoff relevant, understanding the motivations for break aversion is crucial.²³ If misperceptions are persistent—or if correcting misperceptions is costly—an obvious policy suggestion would be mandating breaks. However, in the context of no direct supervisor (e.g., self-employed individuals) this may prove to

23. Our current design does not allow us to draw any behavioral conclusions about how break avoidance would evolve over the course of a work day. However, as proxy we can assume things are static, and we can calculate the monetary implications in the context of an 8h work day. These back of the envelope calculations can be found in appendix ??.

be challenging. In the gig economy, market platforms could explore mandates on maximum work hours.²⁴ A similar misalignment between returns and demand may still arise in managed work settings with piece-rate jobs that are common in lower-income countries. To understand the role and potential of break mandates, it would also be crucial to have data on manager's beliefs and their enforcement. We discuss differences in decision making for oneself versus others in Appendix ??.

24. Enforcement of these may be tricky as well. E.g., The rideshare apps Uber and Lyft allow drivers to work for at most 12 consecutive hours. Afterwards they must take a 6h break. However, there is anecdotal evidence of drivers in the US switching between these two apps instead of resting.

7 Main Text Figures

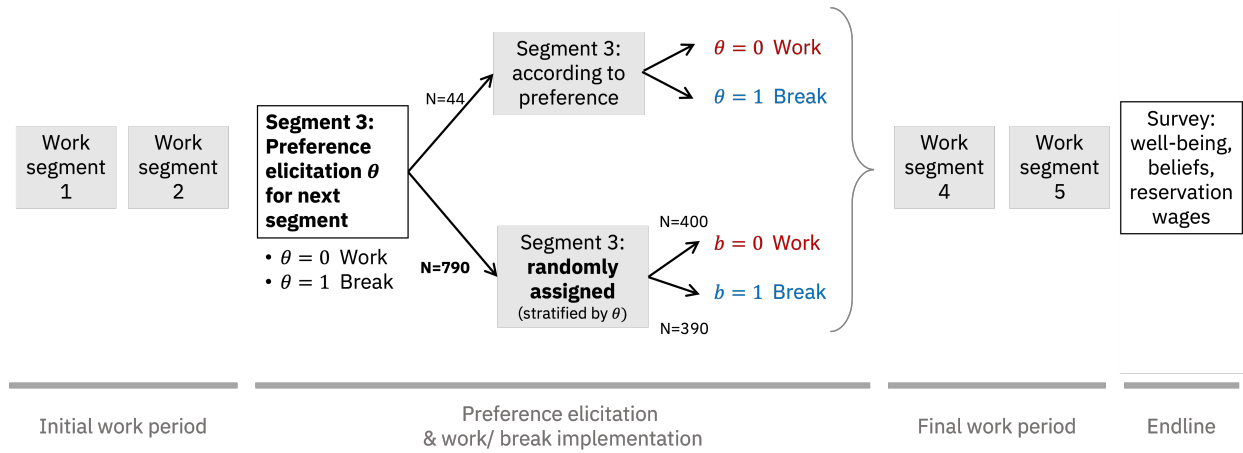


Figure 1: Experimental Design

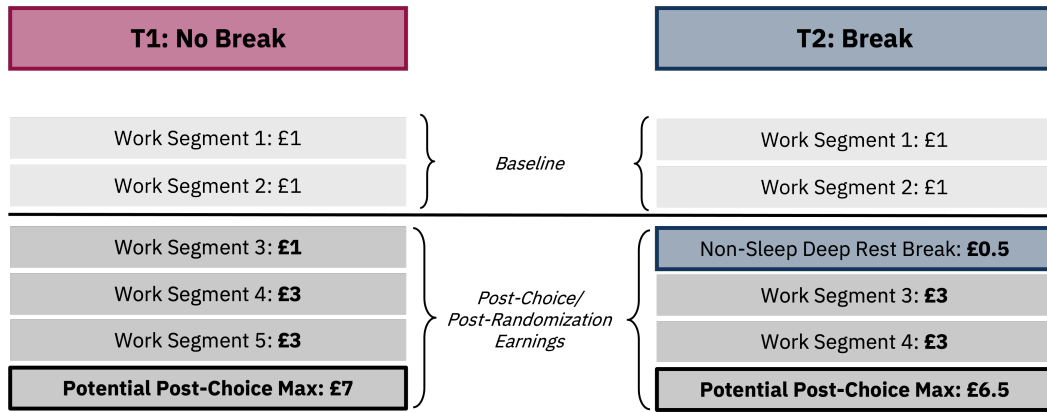


Figure 2: Potential Earnings by Schedule Assignment

Notes: This figure displays the maximum possible earnings that could be achieved in each schedule. During work segments, a bonus payment could be earned if (i) participants correctly identified more than 75% of all flights, and if (ii) they did not miss more than 2 not-safe flights. During the rest segment, participants had to attend to the audio script (which was verified via attention questions afterwards).

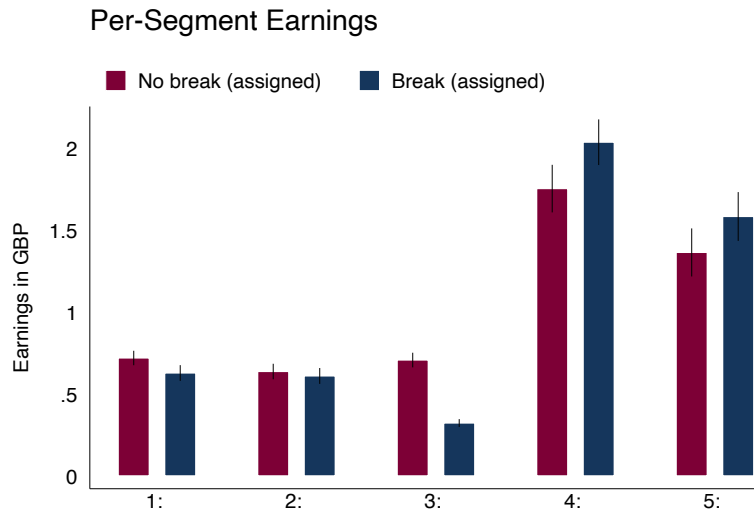


Figure 3: Realized Earnings Per Segment (including break compliance payments)

Notes: Observations restricted to individuals without duplicate entries and that completed the entire work period. Black bars indicated 95% confidence intervals. This figure plots the average realized earnings of workers in a given segment. In segment 1 and 2, workers could earn £1 for every successfully completed segment. In segment 3, workers who were assigned to work could earn £1 if they successfully completed the segment; workers who were assigned to rest could earn £0.5 if they attended to the break audio script. In segment 4 and 5, workers could earn £3 for every successfully completed segment.

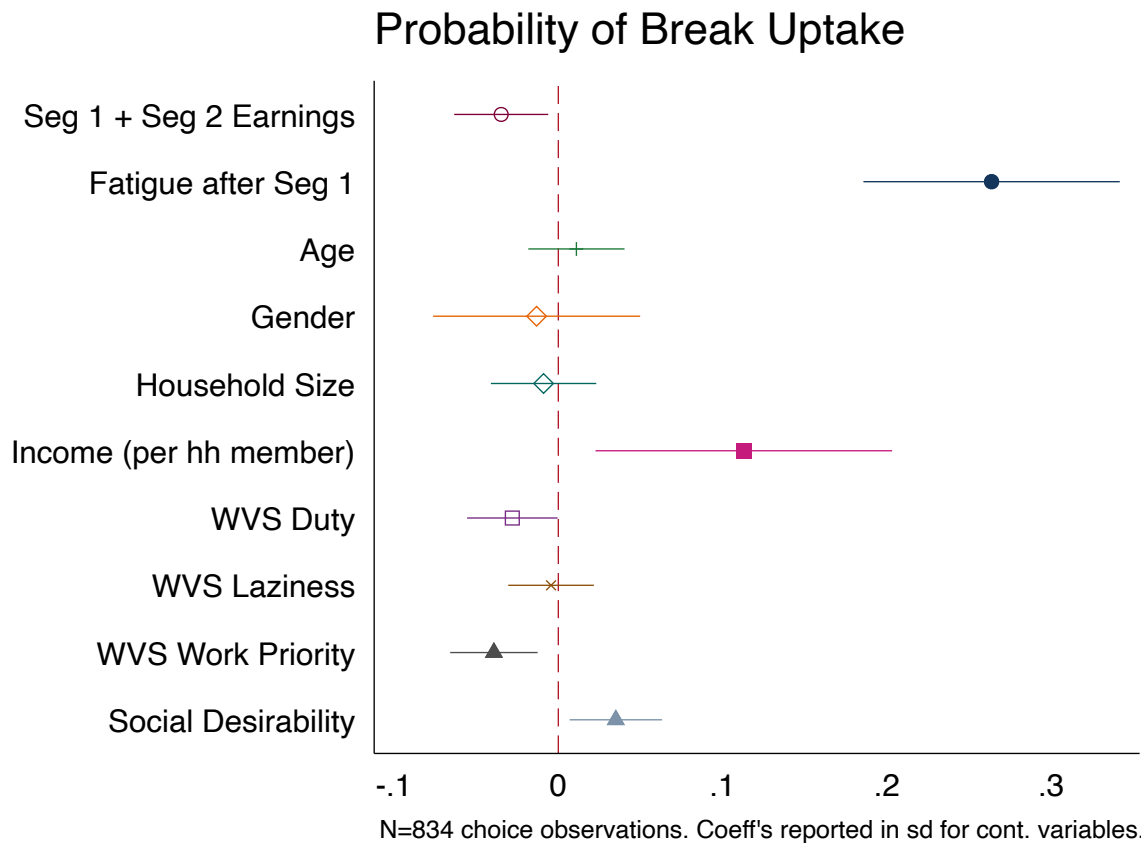


Figure 4: Correlates of Demand for Break in Segment 3

Notes: This figure plots standardized coefficient estimates for bivariate regressions of the preference to take a break on various covariates. Bars represent standard errors. Observations restricted to individuals without duplicate entries and individuals that completed the entire work period.

8 Main Text Tables

Table 1: Balance Table

Variable	(1) No Break (assigned)		(2) Break (assigned)		(1)-(2) Pairwise t-test
	N	Mean/(SE)	N	Mean/(SE)	Mean difference
Social Desirability	400	7.362 (0.097)	390	7.364 (0.100)	-0.002
WVS Laziness	400	0.198 (0.062)	390	0.190 (0.063)	0.008
WVS Duty	400	0.745 (0.049)	390	0.754 (0.048)	-0.009
WVS Work Priority	400	-0.282 (0.061)	390	-0.233 (0.061)	-0.049
Education	371	4.854 (0.050)	363	4.917 (0.048)	-0.063
Age	371	29.588 (0.362)	363	30.512 (0.450)	-0.925
Share African	309	1.000 (0.000)	299	1.000 (0.000)	.n
Gender	371	1.725 (0.024)	363	1.722 (0.024)	0.003
Household Size	371	4.296 (0.078)	363	4.515 (0.097)	-0.219*
Income (per hh member)	371	2826.754 (77.536)	363	2766.905 (79.887)	59.849
Seg 1 Mistakes	400	2.010 (0.126)	390	2.262 (0.129)	-0.252
Seg 2 Mistakes	400	2.723 (0.162)	390	2.859 (0.166)	-0.136

*** p<0.01, ** p<0.05, * p<0.10

Notes: Balance by assignment to break treatment. Observations restricted to individuals without duplicate entries and that completed the entire work period.

Table 2: Post-Break Earnings

	(1)	(2)
Break (assigned)	0.693*** (0.151)	0.721*** (0.159)
Observations	790	734
Constant	0.835	0.832
Seg 1 & 2 Performance	Yes	Yes
Demographics	No	Yes

*** p<0.01, ** p<0.05, * p<0.10

Notes: Robust standard errors in parentheses. Observations restricted to individuals without duplicate entries and that completed the entire work period. Participants without data on demographic control variables are dropped in Column 2. This table regresses total earnings in post-break segments 4 and 5 on whether an individual was randomly assigned a break. All columns control for baseline performance in segment 1 and segment 2, capturing the number of bonuses received by an individual. Demographic controls in Column 2 include age, providence, education level, gender, household size, per-person household income, and employment.

Table 3: Total Earnings (Segment 3-5)

	(1)	(2)	(3)	(4)
Break (assigned)	0.003 (0.156)	0.028 (0.165)	0.322** (0.158)	0.354** (0.167)
Observations	790	734	790	734
BreakPay	Excl.	Excl.	Incl.	Incl.
NoBreakMean	3.828	3.828	3.828	3.828
Performance	Yes	Yes	Yes	Yes
Demographics	No	Yes	No	Yes

*** p<0.01, ** p<0.05, * p<0.10

Notes: Robust standard errors in parentheses. Observations restricted to individuals without duplicate entries and that completed the entire work period. Participants without data on demographic control variables are dropped in Column 2 and 4. This table regresses total work earnings in segments 3 through 5 on whether an individual was randomly assigned a break. Columns 1 and 2 do not include any payments from the break. For workers assigned to the “no break” group, the total earning outcome variable sums earnings from segment 3, 4, and 5. The maximum possible earnings for workers in the “no break” group are 7. For workers in the “break” group, the total earning outcome variable sums earnings from segment 4 and 5 exclusively. The maximum possible earnings for workers in the “break” group are 6. Columns 3 and 4 do include payments from the break. For all workers, the total earnings outcome variable sums work and/or break bonus earnings from segment 3, 4, and 5. The maximum possible earnings for workers in the “no break” group are 7 and for workers in the “break” group 6.5. All columns control for baseline performance in segment 1 and segment 2, capturing the number of bonuses received by an individual. Demographic controls in Column 2 include age, providence, education level, gender, household size, per-person household income, and employment.

Table 4: Total Earnings by Break Choice **with** Break Compliance Pay (Segment 3-5)

	(1)	(2)	(3)	(4)	(5)	(6)
Break (assigned)	0.306*	0.343*	0.528	0.395	0.281*	0.319*
	(0.169)	(0.181)	(0.403)	(0.459)	(0.169)	(0.181)
Break (chosen)					-0.533	-0.487
					(0.323)	(0.337)
Break (assigned) $\times$ Break (chosen)					0.233	0.215
					(0.438)	(0.455)
Observations	635	585	155	149	790	734
Break Preference	No	No	Yes	Yes	N/A	N/A
Constant	1.267	0.243	1.444	5.553	1.447	1.469
Seg 1 & 2 Performance	Yes	Yes	Yes	Yes	Yes	Yes
Demographics	No	Yes	No	Yes	No	Yes

*** p<0.01, ** p<0.05, * p<0.10

Notes: Robust standard errors in parentheses. Observations restricted to individuals without duplicate entries and that completed the entire work period. Participants without data on demographic control variables are dropped in Column 2, 4, and 6. This table regresses total earnings in segments 3 through 5 on whether an individual was randomly assigned a break. The earnings outcome variable includes compliance payments earned for break attendance among the “break group”. For all workers, the total earnings outcome variable sums work and/or break attendance earnings from segment 3, 4, and 5. The maximum possible earnings for workers in the “no break” group are 7 and for workers in the “break” group 6.5. Columns 1 and 2 restrict the sample to workers who did not want to take a break. Columns 3 and 4 restrict the sample to workers who wanted to take a break. Columns 5 and 6 pool all workers regardless of their break preference. All columns control for baseline performance in segment 1 and segment 2, capturing the number of bonuses received by an individual. Demographic controls in Column 2, 4, and 6 include age, providence, education level, gender, household size, per-person household income, and employment.

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A Model Extension: Random Noise

B Details of Experimental Context

B.1 ITT Assignment of Rest as Lower Bound

C Robustness Checks of the Results

C.1 Relaxing Earnings Constraints

C.2 Analysis of Mistakes